

COMPUTER NETWORKING: CONCEPTS

CSE 3751

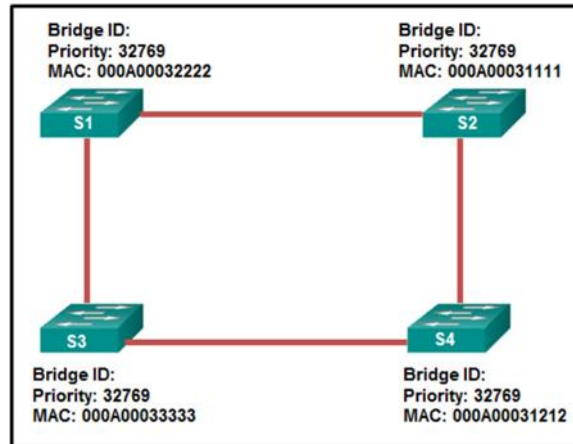
Assignment - 02

1. (a) Explain why cisco 3-tier architecture-based computer network is suitable for a large organisation, highlighting the function of each layer.

(b) In a Cisco 3-tier network architecture, there are 3 access layer switches, each connected to 2 distribution layer switches for redundancy. Additionally, each of the 2 distribution layer switches is connected to 2 core layer switches. Calculate the total number of physical connections required between the access-distribution layers and the distribution-core layers.
2. Name and explain different types of cloud deployment models.
3. With the help of a neat diagram, explain how a proxy server gives protection from hackers while accessing the web servers.
4. Explain why modems are important in internet connectivity from household hosts.
5. A network experiences a loss of 200 packets per second due to congestion. If the packets are transported using UDP, determine how many packets will be retransmitted and justify your answer.
6. What is meant by dynamic routing? State the benefits of using dynamic routing in computer networking.
7. State the importance of AD (Administrative Distance) associated with the routing protocol. If a router is configured with finding a static route and a dynamic route with AD 110 to a destination network, which route will be preferred and why?
8. State the advantage of static routing. Discuss various types of static routes possible between a pair of source and destination nodes.

9. How does STP (Spanning Tree Protocol) help in overcoming the layer-2 loop issue?

Use STP to identify the root bridge and port status in different links for this given network topology.



10. (a) State the meaning of port aggregation and also list out the requirements for port aggregation between two switches.

(b) Given below are the specifications of two switches. Is it possible to make port aggregation between the switches? Justify your answer.

SW1	SW2
Interface: GigabitEthernet Speed: 1000 MB/s Duplex: Half Port mode: Trunk Allowed VLANs: 10, 20 Native VLAN: 99	Interface: GigabitEthernet Speed: 1000 MB/s Duplex: Full Port mode: Trunk Allowed VLANs: 10, 20 Native VLAN: 99

11. Name and discuss various types of VLANs that are supported in computer networking.

12. What is DHCP? Does a wireless router provide DHCP service to host devices? If yes, list the network parameters assigned by the router to the host devices as a part of the DHCP service. If no, justify the reason.

13. Explain the role of beacons in wireless networks. What information is typically included in a beacon?

14. Mention and describe the relevance and importance of the different performance metrics for the operation of any networking device.
15. In a network link, a data packet of size 1500 bytes is sent with transmission speed 10Mbps after a minimum processing time of 0.5 ms at the router. If the link length is 300m and the propagation speed is 2×10^8 m/s, compute the transmission and propagation delay suffered by the packet. Assuming a nil traffic congestion in the link, calculate the minimum one-way delay (i.e. processing time + transmission delay + propagation delay). Later on, if the same packet takes a maximum one-way delay of 10 ms due to traffic congestion, calculate the jitter.
16. Write the default format of a syslog message? Explain each component.
17. What are the different recovery concepts used in any networking organisation for overcoming sudden network disasters? Explain the process of network device backup and restoration in the event of disaster recovery.
18. Define Mean Time To Repair (MTTR) and Mean Time Between Failure (MTBF). If the MTBF of a system is 400 hours and availability is 80%, find the MTTR. [GATE 2018]
19. Assume a critical server operates for 9,000 hours per year and experiences 20 failures within the same year. Further, the IT professionals engaged spend a total of 70 hours in the same year repairing all the failures. Find (i) Mean Time To Repair (ii) Mean Time Between Failure
20. Define fault tolerance of a system with respect to high availability. Name different load balancing/scheduling algorithms and justify how load balancing configuration makes a system fault-tolerant.